



MGM's College of Engineering, Nanded

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

YouTube Channel Performance and Engagement Analytics Dashboard

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Introduction: YouTube Channel Performance and Engagement Analytics Dashboard is a data analytics system designed to collect, analyze, visualize, and compare YouTube channel performance data. The system helps creators understand channel growth, audience engagement, and overall performance using metrics such as subscribers, views, videos, likes, comments, engagement rate, and upload frequency.

System Architecture

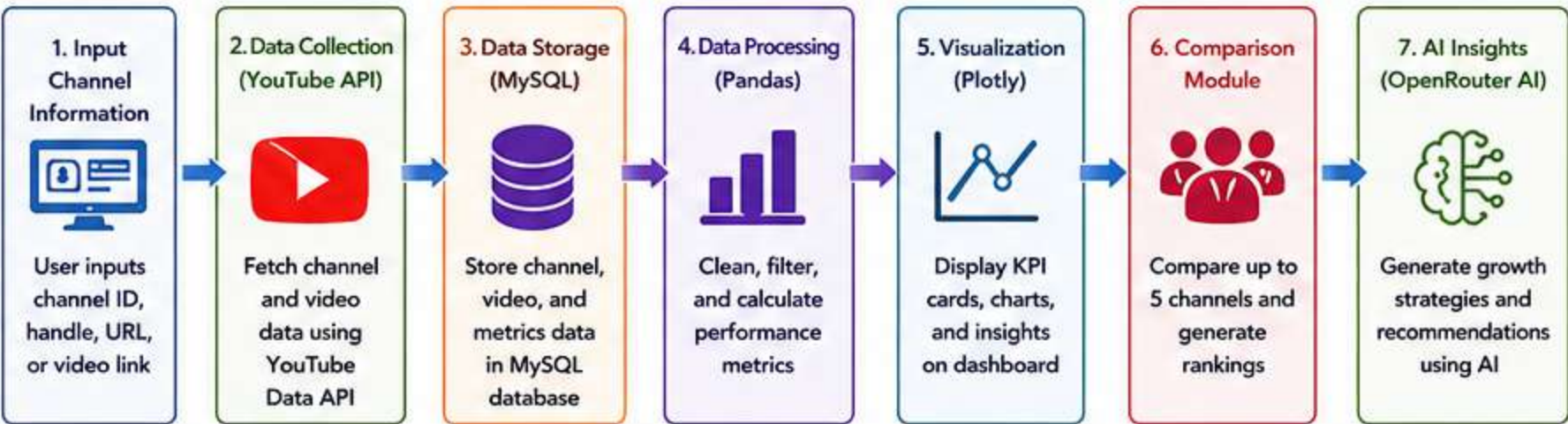
- **Frontend:** Streamlit web user interface
- **Input Module:** Accepts channel ID, handle, channel URL, custom URL, or video link
- **API Layer:** Fetches public channel and video data using YouTube Data API
- **Database:** Stores channel, video, and metric data in MySQL
- **Processing Module:** Uses Pandas for data cleaning, filtering, and metric calculation
- **Visualization:** Displays charts and KPI cards using Plotly
- **AI Module:** Generates growth strategies using OpenRouter AI API

Key Features

- Channel performance dashboard
- Subscriber, views, and video analysis
- Engagement rate calculation
- Subscriber-to-view ratio analysis
- Upload frequency analysis
- Top video performance analysis
- Channel comparison up to 5 channels
- Leaderboard ranking and benchmark analysis
- AI-generated growth strategy recommendations

Methodology

1. User enters YouTube channel link, channel ID, or video link
2. System resolves and fetches channel details
3. Data is collected using YouTube Data API
4. Data is stored in MySQL database
5. Pandas processes and calculates performance metrics
6. Dashboard displays visual insights using Plotly charts
7. AI module generates channel growth recommendations



Conclusion: The system provides an interactive and useful solution for analyzing YouTube channel performance and engagement. It combines Python, Streamlit, MySQL, Pandas, Plotly, YouTube Data API, and OpenRouter AI API to help users understand channel growth and make better content improvement decisions.

Mapping of PO's

- **PO2:** Problem Analysis
- **PO3:** Design/Development of Solutions
- **PO5:** Modern Tool Usage
- **PO11:** Project Management

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